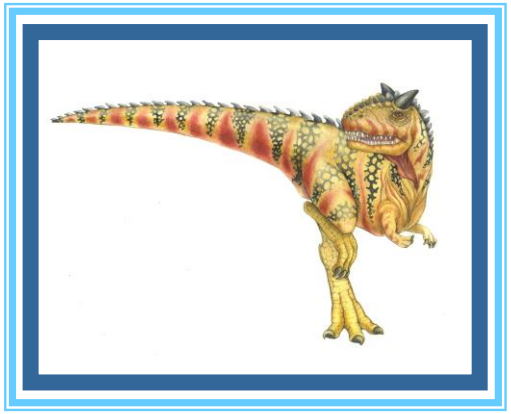
Chapter 7: Synchronization Examples



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Chapter 7: Synchronization Examples

Classic Problems of Synchronization
Synchronization within the Kernel
POSIX Synchronization
Synchronization in Java
Alternative Approaches

Classical Problems of Synchronization

Classical problems used to test newly-proposed synchronization schemes

- Bounded-Buffer Problem
- Readers and Writers Problem
- Dining-Philosophers Problem

Bounded-Buffer Problem

n buffers, each can hold one item

Semaphore **`mutex`** initialized to the value 1

Semaphore **`full`** initialized to the value 0

Semaphore **`empty`** initialized to the value n

Bounded Buffer Problem (Cont.)

The structure of the producer process

```
do {
      ...
       /* produce an item in next_produced */
      ...
    wait(empty);
    wait(mutex);
       ...
       /* add next produced to the buffer */
       ...
    signal(mutex);
    signal(full);
 } while (true);
```

Bounded Buffer Problem (Cont.)

The structure of the consumer process

```
Do {
    wait(full);
    wait(mutex);
        ...
    /* remove an item from buffer to next_consumed */
        ...
    signal(mutex);
    signal(empty);
        ...
    /* consume the item in next consumed */
        ...
 } while (true);
```

Readers-Writers Problem

A data set is shared among a number of concurrent processes

- Readers – only read the data set; they do ***not*** perform any updates
- Writers – can both read and write

Problem – allow multiple readers to read at the same time

- Only one single writer can access the shared data at the same time

Several variations of how readers and writers are considered – all involve some form of priorities

Shared Data

- Data set
- Semaphore **`rw_mutex`** initialized to 1
- Semaphore **`mutex`** initialized to 1
- Integer **`read_count`** initialized to 0

Readers-Writers Problem (Cont.)

The structure of a writer process

```
do {
      wait(rw_mutex);
         ...
      /* writing is performed */
         ...
   signal(rw_mutex);
} while (true);
```

Readers-Writers Problem (Cont.)

The structure of a reader process

```
do {
        wait(mutex);
        read_count++;
        if (read_count == 1)
        wait(rw_mutex);
     signal(mutex);
        ...
        /* reading is performed */
        ...
     wait(mutex);
        read count--;
        if (read_count == 0)
     signal(rw_mutex);
     signal(mutex);
} while (true);
```

Readers-Writers Problem Variations

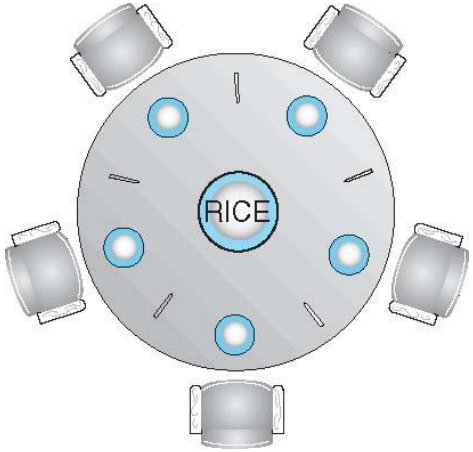
First variation – no reader kept waiting unless writer has permission to use shared object

Second variation – once writer is ready, it performs the write ASAP

Both may have starvation leading to even more variations

Problem is solved on some systems by kernel providing reader-writer locks

Dining-Philosophers Problem



Philosophers spend their lives alternating thinking and eating

Don't interact with their neighbors, occasionally try to pick up 2 chopsticks (one at a time) to eat from bowl

Need both to eat, then release both when done

In the case of 5 philosophers

Shared data

- Bowl of rice (data set)
- Semaphore chopstick [5] initialized to 1

Dining-Philosophers Problem Algorithm

The structure of Philosopher *i*:

```
do {
    wait (chopstick[i] );
     wait (chopStick[ (i + 1) % 5] );

                //  eat

     signal (chopstick[i] );
     signal (chopstick[ (i + 1) % 5] );

                 //  think

} while (TRUE);
```

What is the problem with this algorithm?

Monitor Solution to Dining Philosophers

```
monitor DiningPhilosophers
{
   enum { THINKING; HUNGRY, EATING) state [5] ;
   condition self [5];

   void pickup (int i) {
          state[i] = HUNGRY;
          test(i);
          if (state[i] != EATING) self[i].wait;
}

   void putdown (int i) {
          state[i] = THINKING;
                   // test left and right neighbors
           test((i + 4) % 5);
           test((i + 1) % 5);
}
```

Solution to Dining Philosophers (Cont.)

```
    void test (int i) {
            if ((state[(i + 4) % 5] != EATING) &&
            (state[i] == HUNGRY) &&
            (state[(i + 1) % 5] != EATING) ) {
                 state[i] = EATING ;
             self[i].signal () ;
            }
    }

        initialization_code() {
           for (int i = 0; i < 5; i++)
           state[i] = THINKING;
         }
}
```

Solution to Dining Philosophers (Cont.)

Each philosopher *i* invokes the operations **`pickup()`** and **`putdown()`** in the following sequence:

```
DiningPhilosophers.pickup(i);

    EAT

DiningPhilosophers.putdown(i);
```

No deadlock, but starvation is possible

A Monitor to Allocate Single Resource

```
monitor ResourceAllocator
{
   boolean busy;
   condition x;
   void acquire(int time) {
           if (busy)
              x.wait(time);
           busy = TRUE;
   }
   void release() {
           busy = FALSE;
           x.signal();
   }
initialization code() {
    busy = FALSE;
   }
}
```

Synchronization Examples

Solaris

Windows

Linux

Pthreads

Solaris Synchronization

- Implements a variety of locks to support multitasking, multithreading (including real-time threads), and multiprocessing
- Uses **adaptive mutexes** for efficiency when protecting data from short code segments
  - Starts as a standard semaphore spin-lock
  - If lock held, and by a thread running on another CPU, spins
  - If lock held by non-run-state thread, block and sleep waiting for signal of lock being released
- Uses **condition variables**
- Uses **readers-writers** locks when longer sections of code need access to data
- Uses **turnstiles** to order the list of threads waiting to acquire either an adaptive mutex or reader-writer lock
  - Turnstiles are per-lock-holding-thread, not per-object
- Priority-inheritance per-turnstile gives the running thread the highest of the priorities of the threads in its turnstile

Windows Synchronization

Uses interrupt masks to protect access to global resources on uniprocessor systems

Uses **spinlocks** on multiprocessor systems

Spinlocking-thread will never be preempted

Also provides **dispatcher objects** user-land which may act mutexes, semaphores, events, and timers

Events

- An event acts much like a condition variable

Timers notify one or more thread when time expired

Dispatcher objects either **signaled-state** (object available) or **non-signaled state** (thread will block)

Linux Synchronization

Linux:

- Prior to kernel Version 2.6, disables interrupts to implement short critical sections
- Version 2.6 and later, fully preemptive

Linux provides:

- Semaphores
- atomic integers
- spinlocks
- reader-writer versions of both

On single-cpu system, spinlocks replaced by enabling and disabling kernel preemption

Pthreads Synchronization

Pthreads API is OS-independent

It provides:

- mutex locks
- condition variable

Non-portable extensions include:

- read-write locks
- spinlocks

Alternative Approaches

Transactional Memory

OpenMP

Functional Programming Languages

Transactional Memory

A **memory transaction** is a sequence of read-write operations to memory that are performed atomically.

```
void update()
 {
      /* read/write memory */
  }
```

OpenMP

OpenMP is a set of compiler directives and API that support parallel progamming.

```
void update(int value)
 {
        #pragma omp critical
        {
                count += value
        }
  }
```

The code contained within the **`#pragma omp critical`** directive is treated as a critical section and performed atomically.

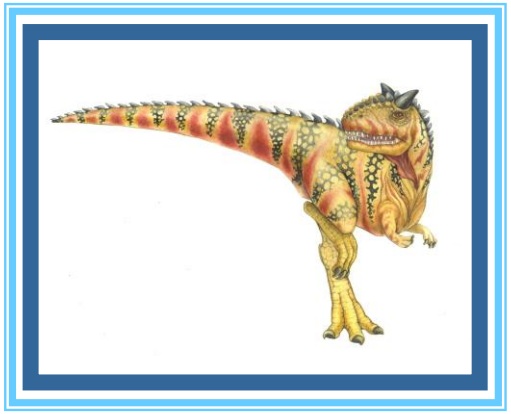
Functional Programming Languages

Functional programming languages offer a different paradigm than procedural languages in that they do not maintain state.

Variables are treated as immutable and cannot change state once they have been assigned a value.

There is increasing interest in functional languages such as Erlang and Scala for their approach in handling data races.

End of Chapter 7



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